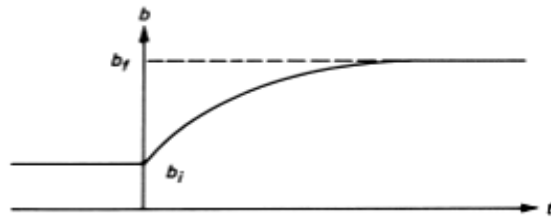
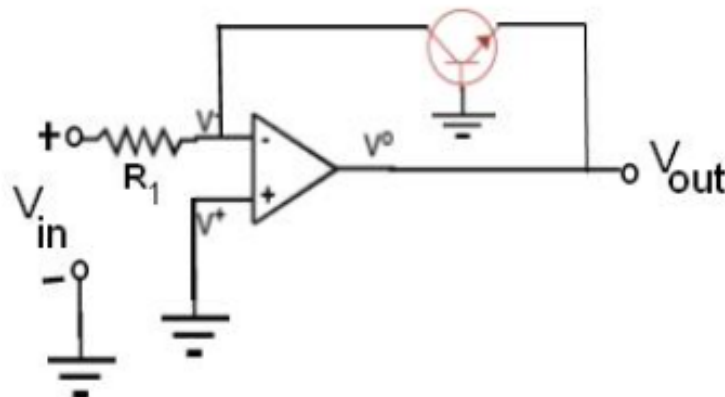
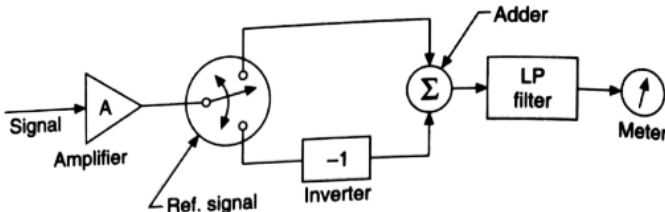
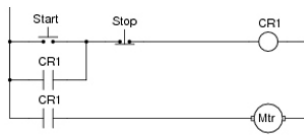
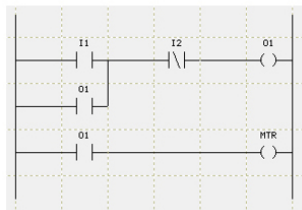
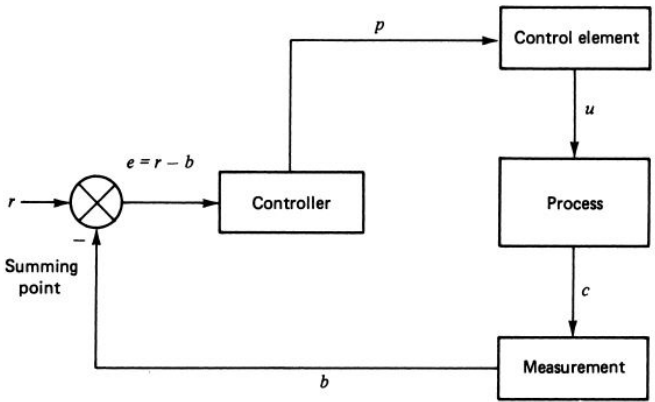
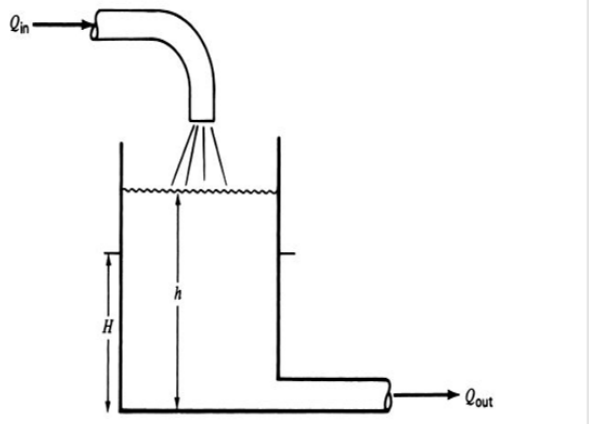


APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY EIGHTH SEMESTER B.TECH DEGREE EXAMINATION, JUNE 2023(2019 Scheme)			
Course Code: EET468			
Course Name: INDUSTRIAL INSTRUMENTATION AND AUTOMATION			
Max. Marks: 100		Duration: 3 Hours	
PART A			
	Answer all questions, each carries 3 marks.		Marks
1	Sensor-time response Note that at the input to the sensor is suddenly changed from an initial value , C_i , to a final value C_f . If the sensor were perfect, its output would be determined by the static transfer function to be b_i before $t=0$ and b_f after $t=0$. However, all sensors exhibit some lag between the output and the input and some characteristic variation in time before settling on the final value. Time response is determined by the solution of a first-order differential equation. A general equation can be written for this response independent of the sensor, the variable being measured, or the static transfer function. 		(3)
2	Any three factors		(3)
3	Operation of a charge amplifier 		(3)
4	Principle and operation of phase sensitive detector.		(3)

	 • A phase sensitive Detector is used to detection of measurement of the amplitude and phase of a periodic signal with respect to another reference signal of the same frequency • The reference signal is assumed to be a square wave driving the switch towards the contactor 1 during the positive half cycle and to contactor 2 during the negative half cycle • The sinusoidal test signal is amplified to assume reasonable amplitude and then applied to the switching system	
5	Process control networks are used to monitor and control industrial processes, such as the production of chemicals or the generation of electricity. They typically consist of a combination of hardware and software components, including sensors, programmable logic controllers (human machine interfaces). • The primary goal of a process control network is to ensure the safe and efficient operation of critical infrastructure. This requires strict control over the flow of data and the access to control systems, in order to prevent unauthorized access or malicious attacks. • Process control networks are typically isolated from other types of computer networks, such as corporate IT networks or the Internet. This isolation helps to prevent external threats from compromising the integrity of the network.	(3)
6	Any three advantages of virtual instrumentation systems Advantages • Lower cost of instrumentation • Easy-to-use graphical user interface • Portability between various computer platforms • Increases the utility of computer • Flexibility	(3)
7	Significance of timers and counters in PLC 1. Variety of time base is available 2. The most common time bases are 0.01 sec, 0.1 sec and 1 sec 3. Accumulator value(ACC)- This is the time that has elapsed, since the timer was last reset. 4. When enabled, a timer updates this continuously 5. Preset Value(PRV)- This specifies the value that the timer must reach before the controller sets the done bit Counter PLC counter utilizes a sensor to count operations, which is processed by software	(3)

	execution in the PLC. Thus the failure rate is reduced and the accuracy level is increased in a PLC counter. The major difference between the counter and the timer is that timer instructions will continually increment its accumulative value at a rate determined by the time base when the enable contact is on	
8	ladder programming 1. Rails- These are vertical lines and provide the sources of energy to relays and logic system 2. Rungs- These are horizontal and contains the branches ,inputs and outputs 3. Branches 4. Inputs 5. Outputs 6. Timer Counter Motor Control PLC Ladder Logic  Motor Control Relay Logic 	(3)
9	Role of MTU in SCADA communication RTUs are designed to operate in harsh environments and are typically installed in remote locations such as oil and gas pipelines, water treatment plants, and power generation facilities. They are capable of collecting data from a variety of analog and digital sensors, such as pressure, temperature, flow rate, and level sensors. RTUs typically have multiple communication ports that allow them to connect to a variety of devices and networks. They use a variety of communication protocols such as Modbus, DNP3, and IEC 61850 to communicate with the central control system and other devices in the field.	(3)
10	Concept of Human Machine Interface in DCS The primary function of the HMI in a SCADA system is to provide a visual representation of the process being monitored and controlled. This is typically done through graphical displays, such as process diagrams, trend charts, and alarm screens. The HMI can also provide real-time status information, allowing operators to quickly identify issues and take corrective action. In addition to displaying process information, the HMI can also be used to control the process. Operators can use the HMI to initiate control actions, such as starting or stopping equipment, adjusting setpoints, and changing control modes. The HMI can also be used to acknowledge alarms and take corrective actions.	(3)
PART B		

Answer one full question from each module, each carries 14 marks.

		Module I	
11	a)	 FIGURE 5 This block diagram of a control loop defines all the basic elements and signals involved.	(8)
	b)	self-regulation, human aided control and automatic control in a process. -2marks each  FIGURE 1 The objective is to regulate the level of liquid in the tank, h, to the value H. ▶ This process has a property called self-regulation ▶ This means that for some input flow rate, the liquid height will rise until it reaches a height for which the output flow rate matches the input flow rate. ▶ Suppose we want to maintain the level at some particular value, H, in Figure 1, regardless of the input flow rate. Then something more than self-regulation is needed	(6)

- ▶ Now the height can be regulated apart from the input flow rate using the following strategy:
- ▶ The human measures the height in the sight tube and compares the value to the set point
- ▶ If the measured value is larger, the human opens the valve a little to let the flow out increase, and thus the level lowers toward the set point
- ▶ If the measured value is smaller than the set point, the human closes the valve a little to decrease the flow out and allow the level to rise toward the set point
- ▶ By a succession of incremental opening and closing of the valve, the human can bring the level to the set point value, H

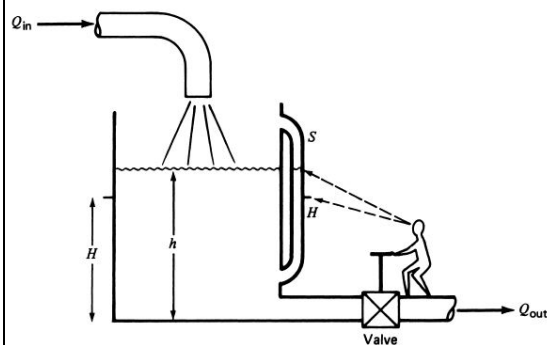
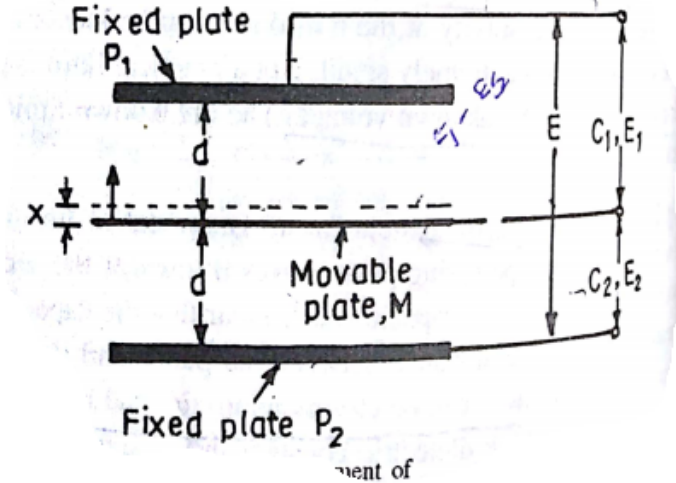
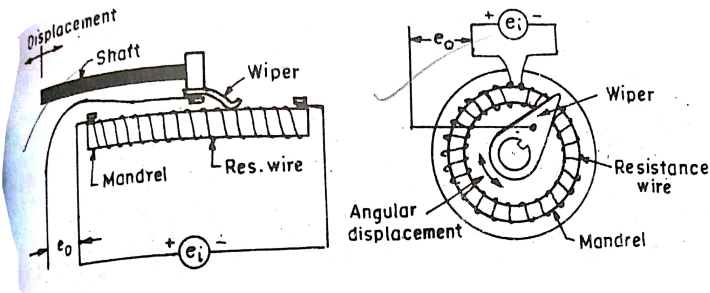
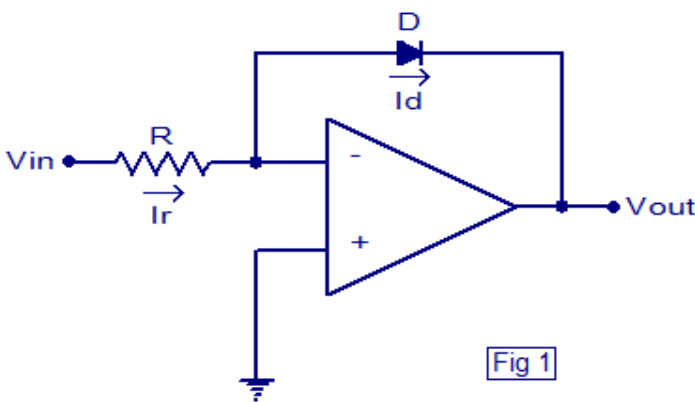


FIGURE 2

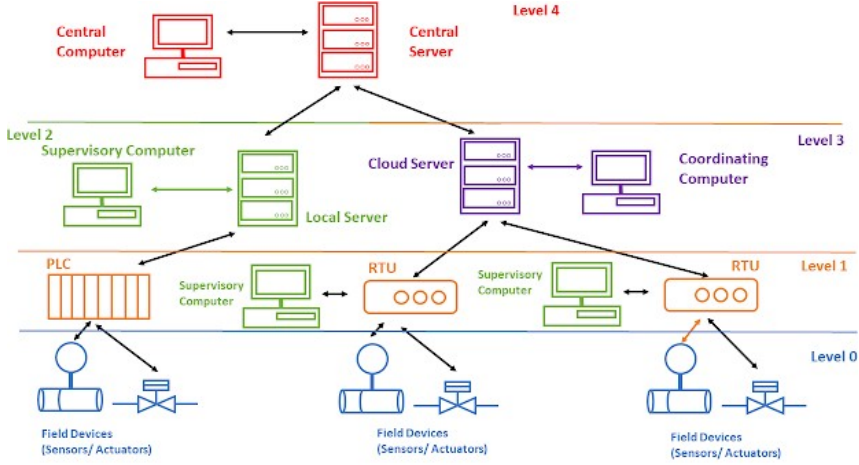
A human can regulate the level using a sight tube, S , to compare the level, h , to the objective, H , and adjust a valve to change the level.

OR

12	a) An AC voltage E is applied across plates P_1 and P_2 and the difference of the voltage across the two capacitances is measured. When the movable plate is midway between the two fixed plates $C_1=C_2$ and therefore $E_1=E_2=E/2$ Voltage across C_1 is $E_1 = \frac{EC_2}{C_1 + C_2} = \frac{E}{2}$ 	(7)
b)	- Any method of varying one of the quantities involved in the above relationship can be the design basis of an electrical resistive transducer. - Can be used for measurement of displacement, force, and pressure - A resistive potentiometer used for the purpose of voltage division is called a POT. - It consists of a resistive element provided with a sliding contact. The sliding contact is called a wiper. The motion of the sliding contact may be translatory or rotational - The resistive element of the POT may be excited by either DC or AC voltage - The POT is a passive transducer since it requires an external power for its operation - The resistive body of potentiometer may be wire wound - The resistive elements are also made up from cermet, hot moulded carbon, carbon film and thin metal 	(7)
	Module II	

13	a)	 Fig 1 Opamp - diode log amplifier $V_{out} = K \ln \left(\frac{V_{in}}{V_{ref}} \right)$	(8)
	b)	Differential amplifier circuit diagram -3 marks Explanation-3 Marks	(6)
		OR	
14	a)	Electrical Actuators explanation	(8)
	b)	Fluid valves explanation	(6)
		Module III	
15	a)	Fibre optic communication Figure-4 marks Explanation-4 Marks	(8)
	b)	Comparison any six points	(6)
		OR	
16	a)	Pneumatic transmission explanation	(8)
	b)	Comparison any six points	(6)
		Module IV	
17	a)	Counter explanation-4 marks Example-4 marks	(8)
	b)	Importance of power supply in PLC explanation	(6)
		OR	

18	a)	<table><tr><th colspan="3">AND</th></tr><tr><th>Input A</th><th>Input B</th><th>Out X</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table> A B X <table><tr><th colspan="3">XOR</th></tr><tr><th>Input A</th><th>Input B</th><th>Out X</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table> A B X A B X A B X <tr><td></td><td>b)</td><td> Importance of CPU in PLC Computer Center Coordinating Computer Supervisory Computers μC Plant Coordinating Computer Supervisory Computers μC Plant Level 4 Production Scheduling Level 3 Production Control Level 2 Plant Supervisory Level 1 Direct Control Level 0 Field Level <tr><td></td><td></td><td>Module V</td></tr></td></tr>	AND			Input A	Input B	Out X	0	0	0	0	1	0	1	0	0	1	1	1	XOR			Input A	Input B	Out X	0	0	0	0	1	1	1	0	1	1	1	0		b)	Importance of CPU in PLC Computer Center Coordinating Computer Supervisory Computers μC Plant Coordinating Computer Supervisory Computers μC Plant Level 4 Production Scheduling Level 3 Production Control Level 2 Plant Supervisory Level 1 Direct Control Level 0 Field Level <tr><td></td><td></td><td>Module V</td></tr>			Module V
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19	a)	Architecture of DCS Figure-4 marks Explanation-4 Marks	(8)
	b)	Main components of SCADA  Figure: Architecture of SCADA System	(6)
		OR	
20	a)	Advantages-3 Marks Disadvantages- 3 marks Applications-2 marks	(8)
	b)	Significance and features of the DNP3 protocols DNP3 (Distributed Network Protocol version 3) is a widely used protocol for communication between devices in SCADA (Supervisory Control and Data Acquisition) systems. The protocol was developed by the Westronic Division of Harris Corporation in the 1990s and has since become an industry standard for communication in critical infrastructure systems. DNP3 is a binary protocol that uses a master/slave architecture for communication between devices. The protocol supports both serial and Ethernet communication and is designed to operate in harsh and noisy environments. DNP3 provides a number of features that make it suitable for use in SCADA systems. These features include: Data integrity: DNP3 provides checksums and error detection mechanisms to ensure the integrity of data transmitted between devices. Time synchronization: DNP3 supports time synchronization between devices, which is critical for maintaining accurate timestamps for data. Scalability: DNP3 supports multi-level hierarchies of devices, allowing for the creation of complex systems with multiple levels of control and supervision. Security: DNP3 includes several security features, such as authentication and encryption, to protect against unauthorized access and data tampering. DNP3 is used in a wide range of industries, including power generation and distribution, water and wastewater management, and oil and gas pipelines. It is	(6)

	known for its reliability and robustness in harsh environments and is widely used in critical infrastructure systems around the world	

